BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 020 · Embedded Systems

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. When can additional CPU cores improve performance?

Q2. What makes an algorithm clear?
A. Steps in any order
B. Missing conditions
C. Instructions that need guessing
D. Precise steps in order

Q3. What does selection do?

Q4. Linear-search [7, 2, 9, 4] for 7, starting at index 0. List the values checked. How many comparisons are needed?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

Embedded system — A computer built into a larger device to perform a specific function.

Dedicated function — A particular task for which a system has been designed.

Sensor — An input device that detects or measures a physical condition.

Actuator — An output device that produces a physical action, such as switching or moving a mechanism.

Real-time operation — A system responds to inputs within a required time limit.

Non-volatile memory — Memory that keeps stored information even when power is switched off.

User interface — The controls or display through which a person interacts with a system.

New learning

Built into a larger device: An embedded system is a computer built into a larger device to perform a specific function. Unlike a general-purpose PC, it is designed for a particular job, such as controlling an automatic device.

Input → process → output: A sensor or switch provides an input. The embedded processor follows stored instructions to make a decision, then controls an output. The output might be a display, motor, light or heater.

Software and memory: The dedicated control program is commonly stored in non-volatile memory (such as ROM/flash) so it remains available when power is off. RAM holds temporary data while the system runs.

Characteristics depend on the job: Embedded systems often have limited user interfaces and are designed to use little power. Some must respond within a fixed time (real-time operation), for example where a delay would affect safety. Not every device has every feature.

Worked example

A washing machine uses a processor to control its wash cycle. Explain why it is an embedded system, identify a sensor input, and name an output. [4 marks]

1. Identify the machine's job: controlling the washing cycle. The controller is part of a larger washing machine, not a general-purpose computer.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

2. It is a computer built into another device [1] designed for the specific job of controlling the wash cycle [1].

3. A water-level sensor sends information to the controller [1]. Other relevant sensors could include temperature sensors.

4. The controller may switch the drum motor on/off as an output [1]. Total: 4 clear marking points — larger device, dedicated purpose, input, output.

We Do · Practise with guidance

Guided 1

A thermostat keeps a classroom warm. Explain why its controller is an embedded system. Name one input and one output. [4 marks]

Worked method: It is a computer built into a larger thermostat/heating-control device [1] for a dedicated temperature-control function [1]. A temperature sensor provides the input [1]; the controller switches the heating on or off as an output [1].

Guided 2

A pedestrian crossing uses a push button and signal lights. Explain why its controller is an embedded system, giving an input and an output. [4 marks]

Worked method: The controller is built into the crossing system [1] and has the dedicated function of controlling the signals [1]. The push button provides an input [1]; the red/green pedestrian lights are an output [1].

Guided 3

A fitness tracker runs from a small battery and has only a tiny display. Explain why low power and a limited user interface are appropriate. [4 marks]

Worked method: Low power consumption [1] allows longer operation between charges [1]. A limited interface [1] is enough to display brief readings and reduces size/complexity for a device with a specific purpose [1].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Change temperature to 27. Does anything print?

```
temperature = 29
if temperature > 28 then
    print("Ventilation on")
endif
```

write

Question 1

State TWO characteristics of an embedded system. [2 marks] Source: Adapted from the Embedded Systems questions in your Paper 1 revision booklet, 1.1.3.

Question 2

A car-park barrier uses a vehicle sensor and a motor to lift the barrier when authorised. Explain why its controller is an embedded system, and identify an input and an output. [4 marks] Source: Original OCR J277-style application question based on specification 1.1.3.

Question 3

A battery-powered smoke alarm uses a sensor to detect smoke and activates a loud sounder when smoke is detected. Explain THREE embedded-system characteristics that make it suitable for this job. [6 marks] Source: Original OCR J277-style applied question based on the characteristics listed in your Paper 1 revision booklet, 1.1.3.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Exit Ticket · Answer before checking

Every embedded system must have a full keyboard and monitor.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. More cores can help when work can be split between them, so tasks can make progress at the same time.

Q2. D — Precise steps in order

Q3. It chooses a branch using a condition. Selection can be nested.

Q4. (a) [7]; 1 comparison(s), found at index 0.

I Do / We Do pages · Guided practice

Q1. It is a computer built into a larger thermostat/heating-control device [1] for a dedicated temperature-control function [1]. A temperature sensor provides the input [1]; the controller switches the heating on or off as an output [1].

Q2. The controller is built into the crossing system [1] and has the dedicated function of controlling the signals [1]. The push button provides an input [1]; the red/green pedestrian lights are an output [1].

Q3. Low power consumption [1] allows longer operation between charges [1]. A limited interface [1] is enough to display brief readings and reduces size/complexity for a device with a specific purpose [1].

You Do pages · Code First

```
temperature = 27
if temperature > 28 then
    print("Ventilation on")
endif
```

Expected result: No output

You Do pages · Independent questions

Q1. Designed for a specific function [1] and built into a larger device [1]. Also acceptable where relevant: a limited user interface, low power consumption, or real-time response. Credit two distinct valid characteristics.

Q2. It is a computer built into the wider barrier system [1] for the specific function of controlling entry [1]. Input: a vehicle-detection sensor [1]. Output: controlling the

motor to raise/lower the barrier [1]. 'Barrier' alone is not a specific input or output.

Q3. Dedicated purpose [1]: designed specifically to detect smoke and raise an alarm [1]. Low power [1]: helps the device run for a long time from its battery [1]. Real-time response [1]: detects smoke and triggers the sounder within the required time, reducing dangerous delay [1]. Alternatives may earn credit if they correctly explain a relevant characteristic, such as non-volatile program storage. Model mark points for original practice, not an official OCR mark scheme.

Exit Ticket · Model answer

False

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: